CO2xN: Porewater Sulfide

Month Year Samples

2025-12-12

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "May"
sample_year = "2025"
Run_Date = as.Date("05/27/2025", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Brenden Blakley"#Code user

Run_notes=""#any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7","Plate_rr")

Raw_plates <- c("Raw Data/20250527_LTREB_H2S_Plate1.csv","Raw Data/20250527_LTREB_H2S_Plate2.csv","Raw Data

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
    Plate = c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7","Plate_rr"),
    Run_Date = c("05/27/2025","05/27/2025","05/27/2025","05/27/2025","05/27/2025","05/27/2025","05/27/2025
  )

  Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%Y"))


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_H2S_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202505.csv"
rerun_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202505_reruns.csv"

#Stds that should be excluded
#stds_to_remove<-data.frame(Plate=c("Plate4","STD2"),IDs=c("Std 4","Std 3"))
```

##Read in metadata and create similar sample IDs for matching to samples

read in data

#Remove High CV Stds

#Check R2 & Remove Standards if Necessary

```
##   X  Plate Curve     R2    Slope Intercept   Run_Date
## 1 1 Plate1   H2S 0.9922 106.3464   -9.8414 2025-05-27
## 2 2 Plate2   H2S 0.9961 110.5220   -8.7349 2025-05-27
## 3 3 Plate3   H2S 0.9935 103.4806   -8.9462 2025-05-27
## 4 4 Plate4   H2S 0.9969 100.5997   -7.7928 2025-05-27
## 5 5 Plate5   H2S 0.9951 106.8267   -9.7185 2025-05-27
## 6 6 Plate6   H2S 0.9969 100.5997   -7.7928 2025-05-27
```

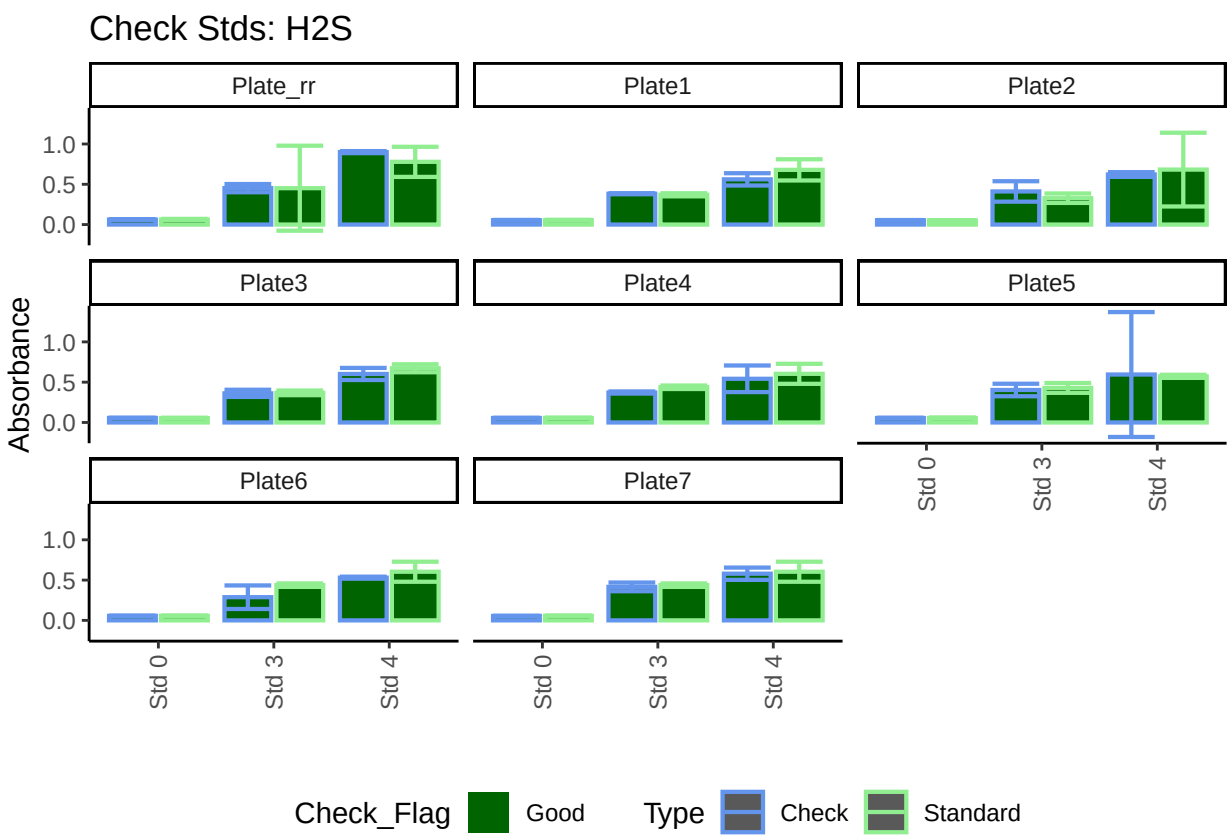
#Return Flags For Standard Curve

```
## [1] "CV Good-Plate1"   "CV Good-Plate2"   "CV Good-Plate3"   "CV Good-Plate4"
## [5] "CV Good-Plate5"   "CV Good-Plate6"   "CV Good-Plate7"   "CV Good-Plate_rr"

## [1] "R2 Good-Plate1"   "R2 Good-Plate2"   "R2 Good-Plate3"   "R2 Good-Plate4"
## [5] "R2 Good-Plate5"   "R2 Good-Plate6"   "R2 Good-Plate7"   "R2 Good-Plate_rr"
```

Remove High CV Check Standards

##Compare Check Standards to Standards



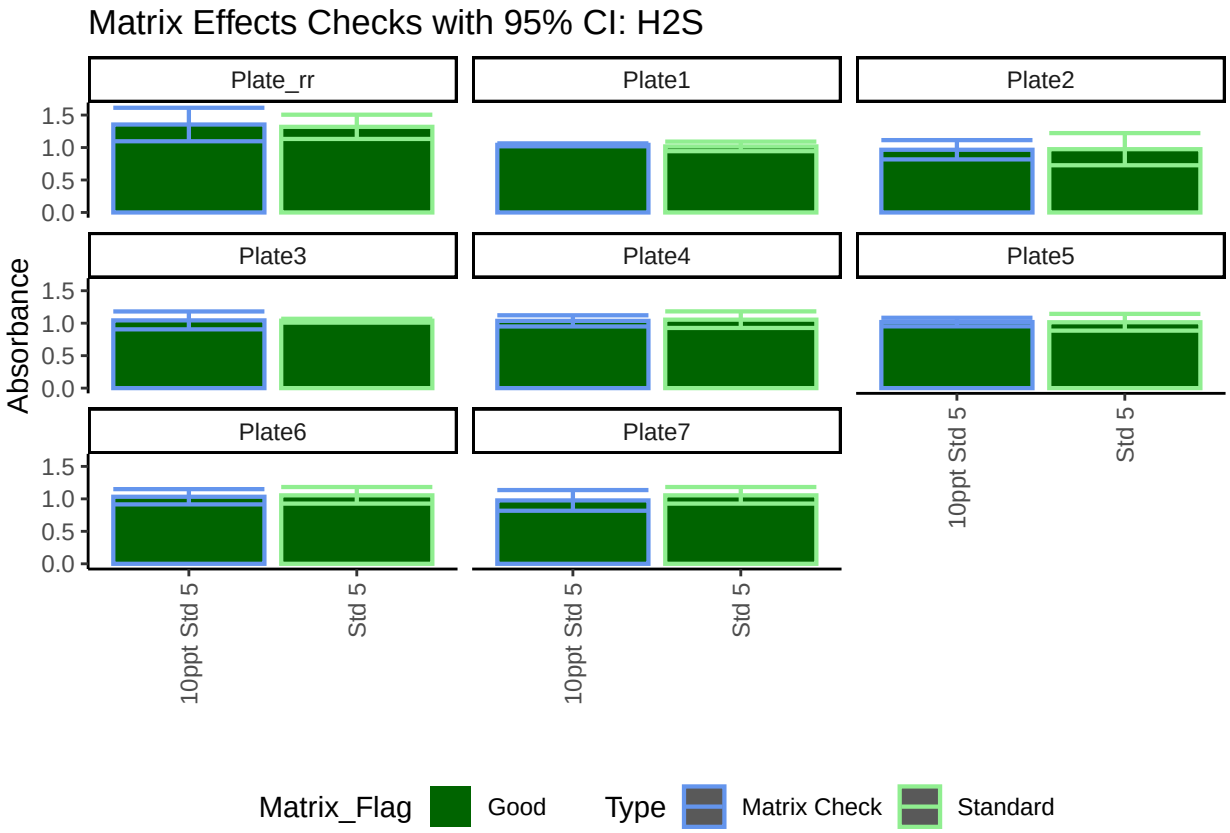
##Display Any Check Flags

```
## [1] "All Checks Good"
```

```
## [1] "Std4Chk_CV_High-Plate5"
```

Remove High CV Matrix

##Compare Matrix to Standards



##Display Any Matrix Flags

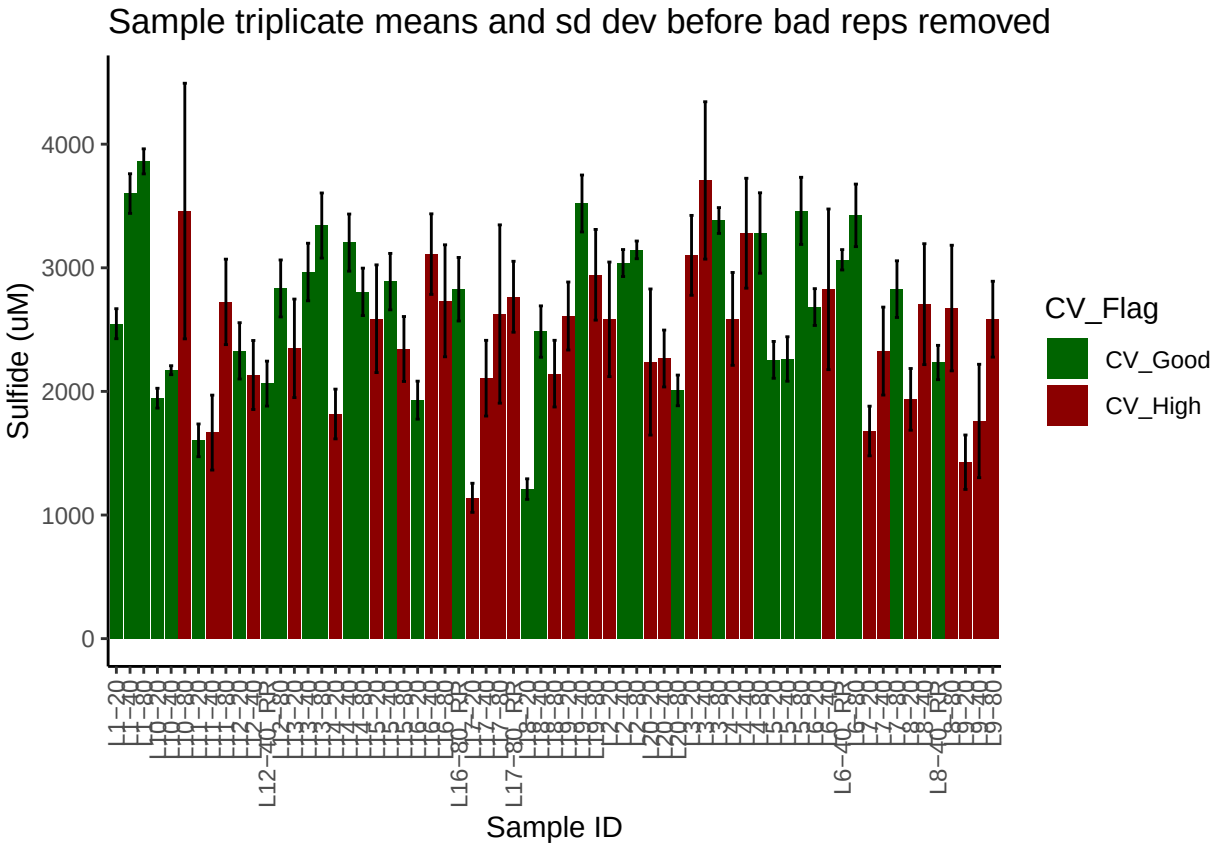
```
## [1] "All Matrix Checks Good"
```

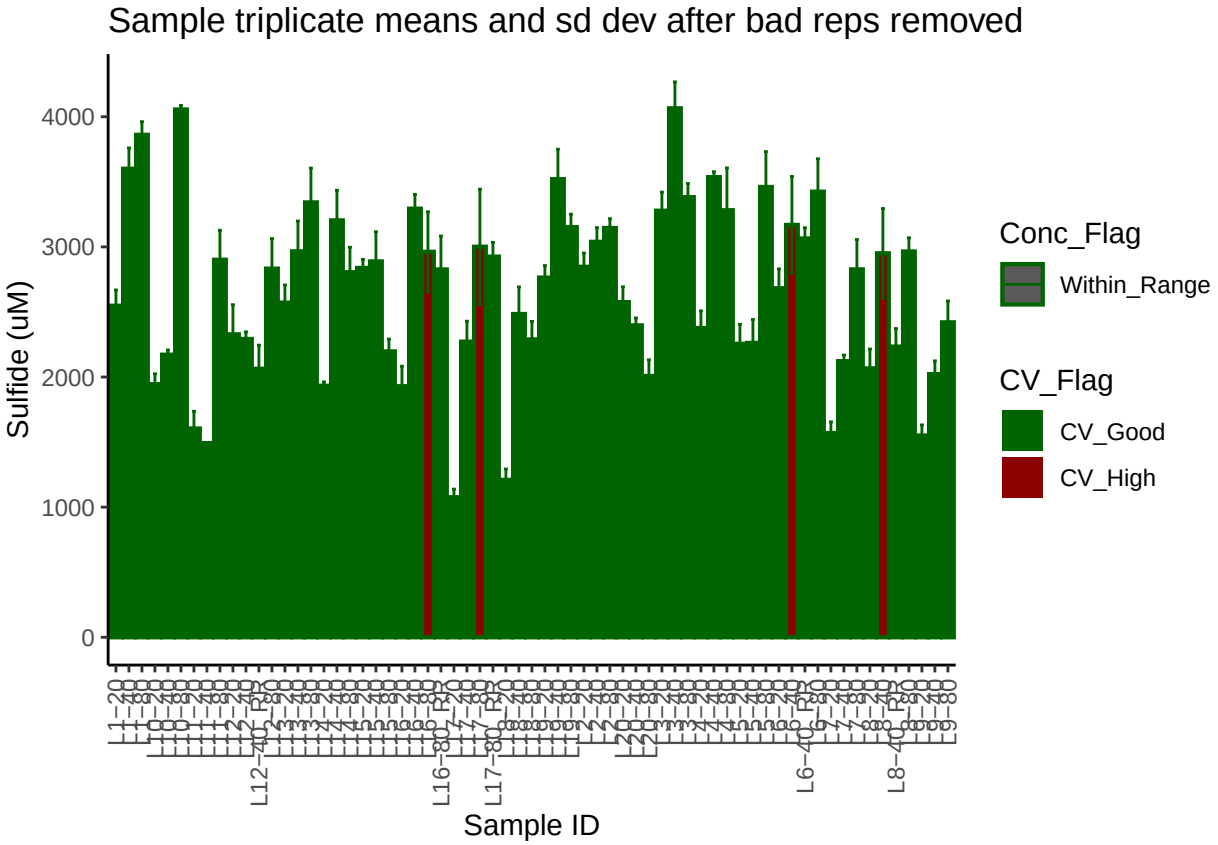
```
## [1] "All Matrix CV Good"
```

Calculate Sulfide Concentrations & Add Concentration Flags

Remove High CV Samples and Average

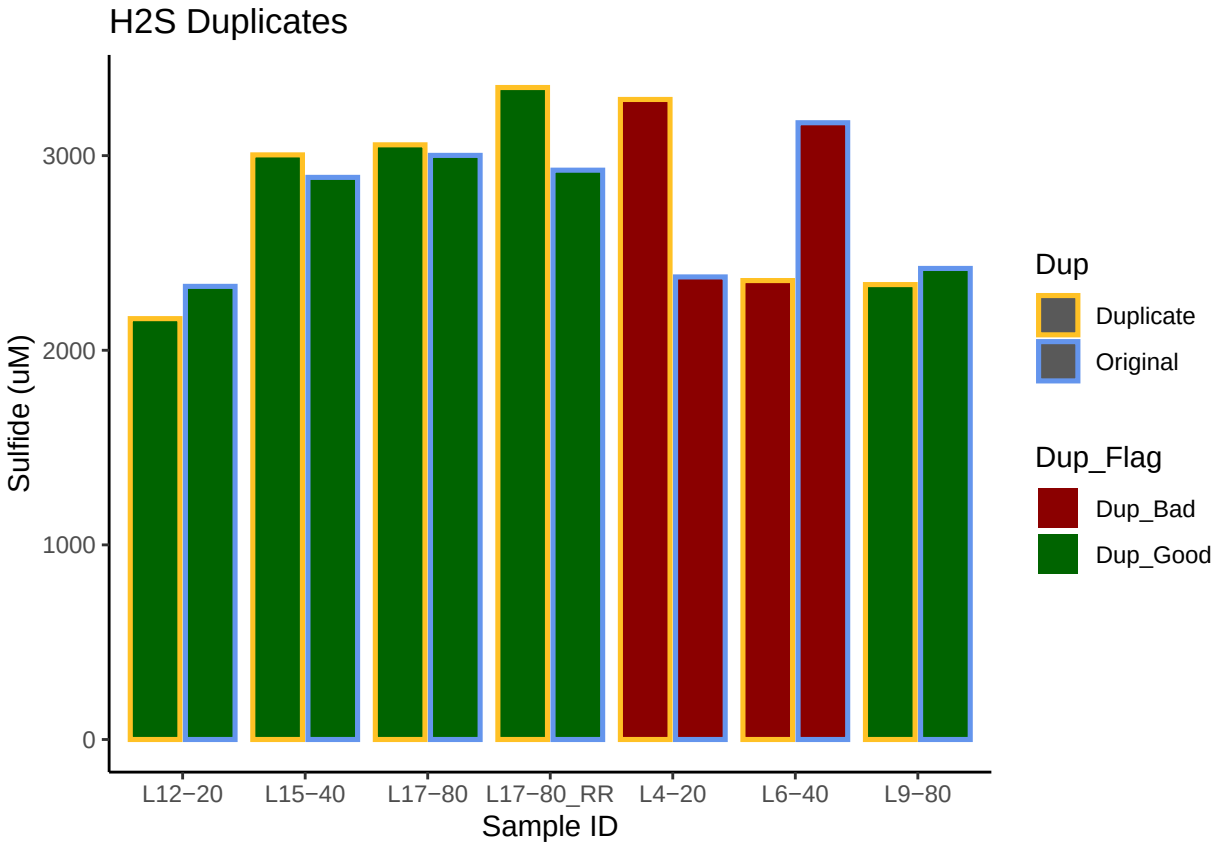
Plot Samples





Check the dups for QAQC

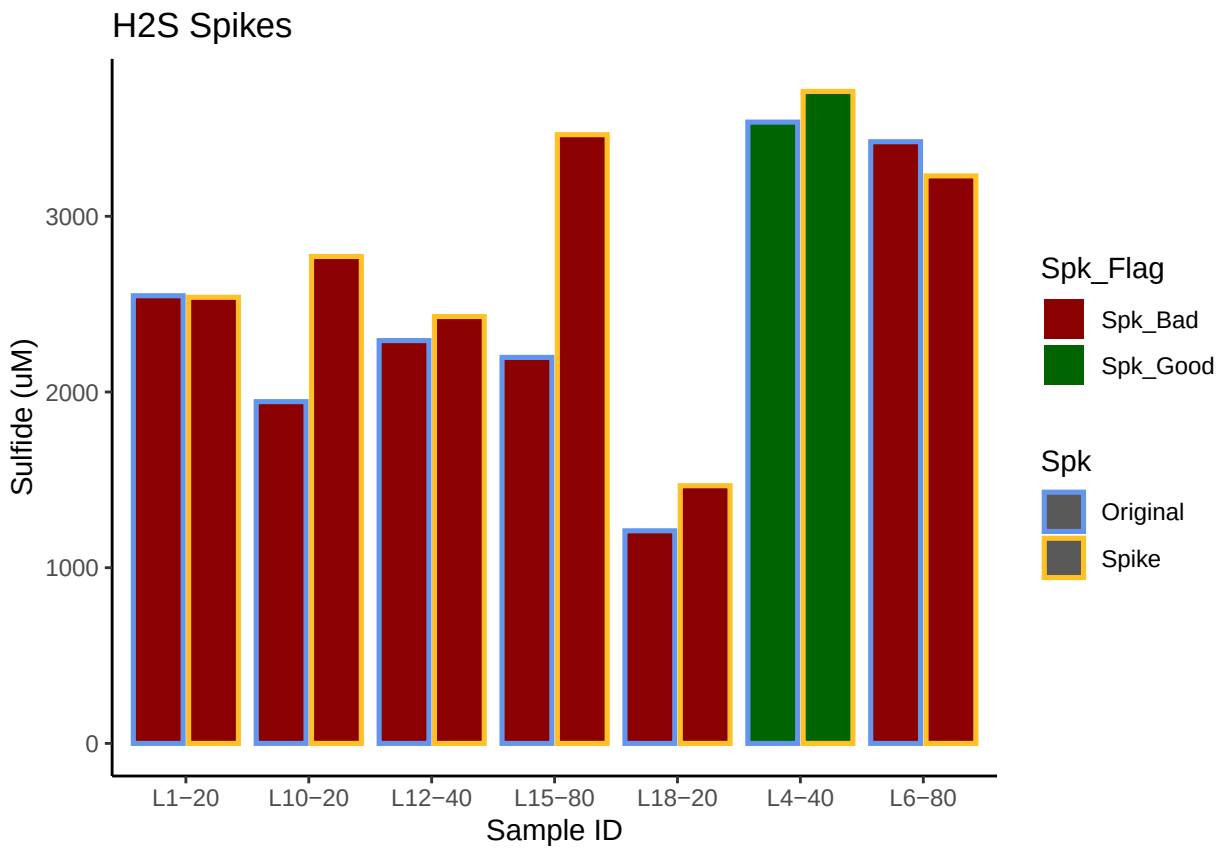
```
## Joining with `by = join_by(Dup_ID)`
```



Return Dup Flags

```
## [1] "Dup_Bad-Plate1"    "Dup_Good-Plate2"   "Dup_Good-Plate3"
## [4] "Dup_Good-Plate_rr"
```

Check the spks for QAQC



##Display Spike Flags

```
## [1] "Spike Missing Pair"
```

```
## [1] "Spk_Bad-Plate1" "Spk_Bad-Plate2" "Spk_Bad-Plate3" "Spk_Bad-Plate4"
```

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

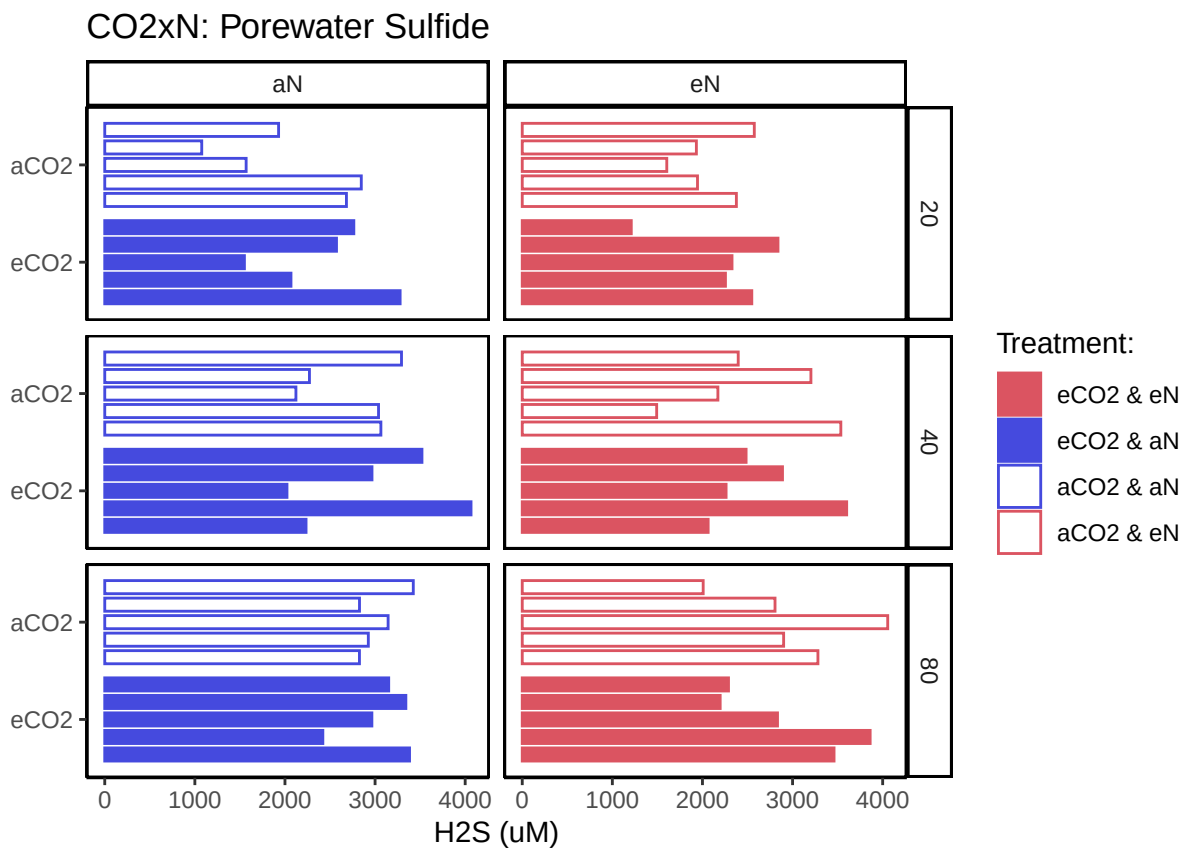
```
## Merge Samples with Metadata
```

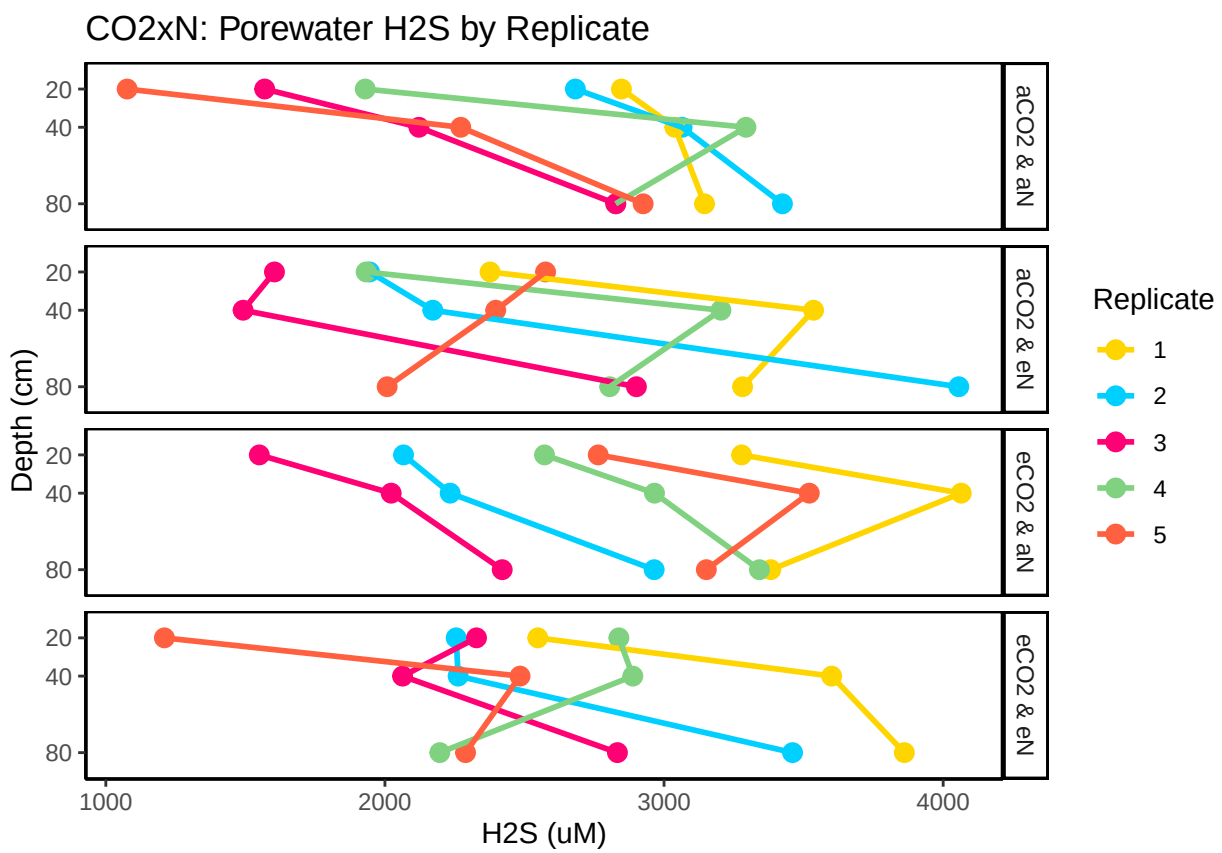
```
## All sample IDs are present in data
```

#Add Flags

#Organize Data

Visualize Data





##Export Data

END